

Instructor: Date:

Learner: Regional Branch:

Program: Center:

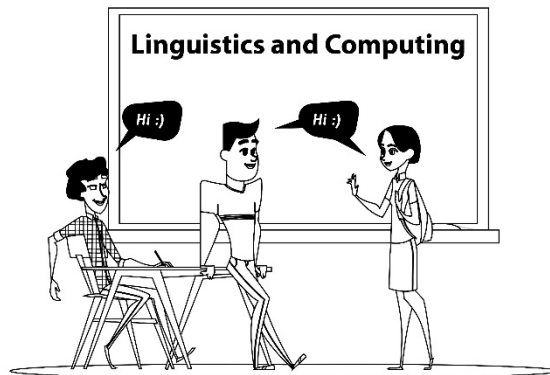
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Let's Start!

Teaching Computers How Humans Talk



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about linguistics and computing.
- I can obtain information about linguistics and computing.
- I can explain the relation between linguistics and computing.

Ask your partner:

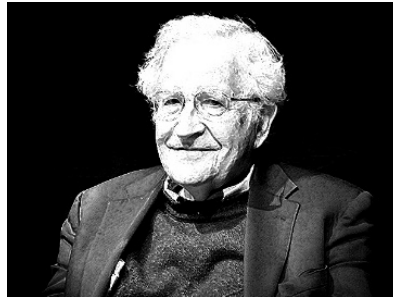
- What are the benefits of having devices that work with voice commands?
- How do you think that programmers make computers recognize human speech?





Let's Explore!

Steven wants to know the relationship between computing and linguistics.



Noam Chomsky

Instructions: Read the text below.

Steven: Teacher, I really appreciate your help with this topic.

Victoria: Don't worry, Steven. So, what questions do you have?

Steven: Well, what do you know about linguistics and computing?

Victoria: Oh, I love talking about that. The first thing you need to know is that there is something called computational linguistics, which studies the structure of natural language and its use as a programming language. The idea of computational linguistics is to involve the creation of a software with linguistic principles.

Steven: That's very interesting, and how did that relationship start?

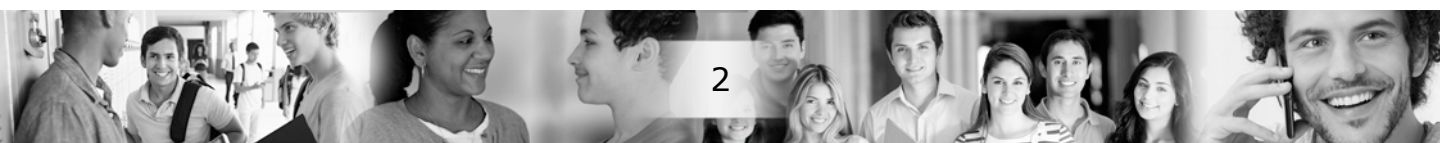
Victoria: The relationship between computing and linguistics started because programmers had problems giving precise and unambiguous definitions for the language structures they used. As a result, they had to look for help in the field of linguistics. When they researched about linguistics, they found that Noam Chomsky could provide them with theories about the structure of languages. Chomsky defined grammar as a group of rules with different levels.

Steven: I'm still a little confused. Why are we talking about Noam Chomsky? I mean, I don't see the connection between language or linguistics and computing yet.

Victoria: It's not really complex, Chomsky contributed to programmers with a concept called "transformational-generative grammar", which is applicable to human languages as a universal structure. Computer scientists use formal grammar rules to define statements in programming languages, consequently, they are able to break down high-level language statements and transform them into low-level instruction codes.

Steven: Really? That's sound incredible. And what are computer scientist doing nowadays with linguistics?

Victoria: I think they are developing new ways to translate languages. As you know, the WWW is evolving very quickly, and programming languages are part of that evolution. Computational linguistics and natural language processing is a big field of study.



**Let's Practice!**

Steven can relate each word with a definition.

Instructions: Match column A with column B.

COLUMN A

1. Linguistics:
2. www:
3. codes:
4. developer:

COLUMN B

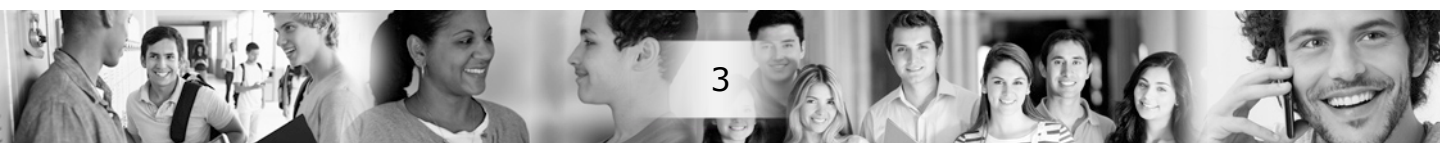
- a. refers to the study of language in terms of phonetics or semantics
- b. set of regulations to design behavior in a system
- c. person that creates computer software
- d. world wide web

**Let's Practice!**

What did Steven learn from his teacher?

Instructions: Circle the correct word/words to complete the text.

Computer scientist needed to find a way to adapt the natural language into a programming language, (consequently / consequence / as result), they used the work of Noam (Shakespeare / Chomsky / Skinner) to develop a theory that allowed them break down high-level language statements and transform them into low-level instruction codes. (Consequence / A result / As a result), CPUs are capable of executing these codes and work with them.





Extension Activity

Linguistics benefits in software programming.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for some information about use of linguistic on programming software.
- Write notes and create a poster where you describe the advantages of using linguistics in computing.
- Share the poster with your partners and answer any question.

Self-Assessment

Yes No

- I look for some information about use of linguistics on programming software.
- I create a poster where I describe the advantages of using linguistics in computing.
- I talk about linguistics on programming software.

☐	☐
☐	☐
☐	☐



Target Words

- linguistics** (noun): refers to the study of language in terms of phonetics or semantics
e.g. The first thing you need to know is that there is something called computational linguistics, which studies the structure of natural language.
- evolve** (verb): to develop in a process
e.g. As you know, the WWW is evolving very quickly, and programming languages are part of that evolution.
- level** (noun): a rank, how something is organized
e.g. Chomsky defined grammar as a group of rules with different levels.
- yet** (adverb): at the present time
e.g. I mean, I don't see the connection between language or linguistics and computing yet.
- break down** (phrasal verb): to separate something into smaller pieces
e.g. They are able to break down high-level language statements and transform them into low-level instruction codes.
- WWW** (abbreviation): world wide web
e.g. As you know, the WWW is evolving very quickly, and programming languages are part of that evolution.



Learning Points

Connectors for Consequence: As a result and Consequently

There are some words that you use to connect two ideas. The first idea represents a cause, and the second idea represents a consequence.

The connectors that help us show a consequence are: **As a result** and **Consequently**.

These two connectors share the same meaning, let's see the next examples:

- *Computer scientists needed help from linguistics, **as a result**, they used the work of Noam Chomsky.*
- *The field of computational linguistics in computer science is very important, **consequently**, the creation of software that can apply linguistics principles is evident.*

Please note that punctuation is very important when you use connectors.

Examples

- *Programmers had problems giving precise and unambiguous definitions for the language structures they used. **As a result**, they had to look for help in the field of linguistics.*
- *Computer scientists use formal grammar rules to define statements in programming languages, **consequently**, they are able to break down high-level language statements and transform them into low-level instruction codes.*

Wh-Hw Questions with Simple Present, Present Continuous, and Simple Past

In order to ask open questions in simple present, present continuous, or simple past, you need to use the **Wh-Hw questions**.

We use these type of questions whenever we demand for specific information about an object, place, date, etc.

Let's review the Wh-Hw questions with the following chart:

What	To obtain information about a topic	What do you think about linguistics in computing?
Where	To obtain information about a place	Where are you putting those codes?





Learning Points

When	To obtain information about a date	When are we going to design our own translation app?
Who	To obtain information about a person	Who was the preeminent linguist?
Which	To obtain information about something already mentioned	Which programming language are we going to use?
Why	To know the cause of something	Why do programmers need to know about linguistics?
Whose	To obtain information about possession	Whose is this computer?
How	To obtain information about a process	How do we program devices to recognize human speech?

Simple Present

You need to use the auxiliary Do (you-we-they) and Does (he-she-it).

Structure: **Wh-Hw** + **do/does** + noun + **verb** + complement + ?

Example: **How do** you **relate** linguistics and programming languages?

Present Continuous

You need to use the verb be (am-is-are) and a verb with (ing).

Structure: **Wh-Hw** + **Verb to be** + noun + **verb (ing)** + complement + ?

Example: **What are** you **implementing** about linguistics in computing?

Simple Past

You need to use the auxiliary **did**.

Structure: **Wh-Hw** + **Did** + noun + **verb** + Complement + ?

Example: **What did** you **read** about linguistics in computing?

Examples

- What questions do you have?
- Why are we talking about Noam Chomsky?
- What are computer scientist doing nowadays with linguistics?